

Helix Thready — Design System

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Rev	Date	Author	Change
1	2026-07-21	swarm (design)	Initial complete draft: token architecture, Thready theme, platform fan-out, typography/spacing/motion, a11y contract, visual-regression testing
2	2026-07-21	swarm (design · review)	Second-pass review: clarified Angular 19 (product) vs 22 (marketing) per Q17; added the mandated Challenges scenario-bank test type to \$8/\$9 ([GAP: 9.3] second half)
3	2026-07-22	swarm (design · Pass 3)	Depth pass: re-verified every token name at source (gh); added the remaining verified core token (--elev-flat) and theme aliases (--fg-2 / --meta / --border-soft); shipped-brand-themes reference table + measured contrasts (helix-green/vasic-red/helix-ota-blue) + the 4-step “add a theme” process (\$3.4); typography weight/variable-axis detail + exact i18n keys
4	2026-07-22	swarm (design · review-fixes)	Rendering fix from the adversarial platform review: the Angular 19/22 [Q17] blockquote had been inserted mid-table in \$7, splitting the platform table so the Desktop/React/KMP/Flutter/TUI rows rendered as a broken second table — moved the blockquote below the (now contiguous, single) table; no row content changed

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1. Position and non/negotiables

The Thready design system is **not a new system**. It is a **brand theme + Thready-specific component set layered on the shared, in-house `basic-digital/design_system`**, which is itself extracted from OpenDesign `[CONSTITUTION §11.4.162]` `[IN-HOUSE: design_system]`. This is mandated:

- **Consume, never fork.** The design system is a dependency; Thready extends it upstream, never vendors or forks it (decision matrix §10.2, `[CONSTITUTION §11.4.28]`).
- **Light + dark are both first-class**, with explicit choice and a system default (\$Frontends, Q-theming) `[OPERATOR]`.
- **One design system for every surface** — Web, Desktop, Mobile, **and TUI** — every widget comes from the design system (\$Frontends: “All widgets MUST BE created from our Design System ... TUI included”).
- **Default theme derived from `Logo.png`** (\$Frontends) — the Thready theme.
- **No invented values** `[CONSTITUTION §11.4.6]`: structural tokens come from the OpenDesign schema defaults; brand color comes from captured pixel evidence of the logo.

Verified status. `design_system` is `FOUNDATION` (real, extracted from HelixOTA, web-only) `[GAP: 8.1 design_system]`. The web/CSS + Angular layer is production-usable today; the non-web arms (`helix_design` for Flutter/Qt, `UI-Components-KMP`) are **scaffolds** and are tracked in §7 and §9. We do **not** claim the non-web variants work.

2. Source: OpenDesign + the shared design_system

OpenDesign (`nexu-io/open-design` , release `0.13.0` at time of writing `[VERIFIED]`) is an agent-driven design-system daemon. Its `brands/engine` derives a token set from brand inputs (logo, seed color, `DESIGN.md`) and its `export.ts` emits the canonical artifacts:

```
DesignTokens — tokensToJson    → tokens.json   (raw, inspectable)
DesignTokens — tokensToCssVars → :root { --... } (CSS custom properties)
SeedToken    — tokensToThemeJson → theme.json   (Ant Design ConfigProvider)
                                     + screenshot-backed PPTX / PDF
```

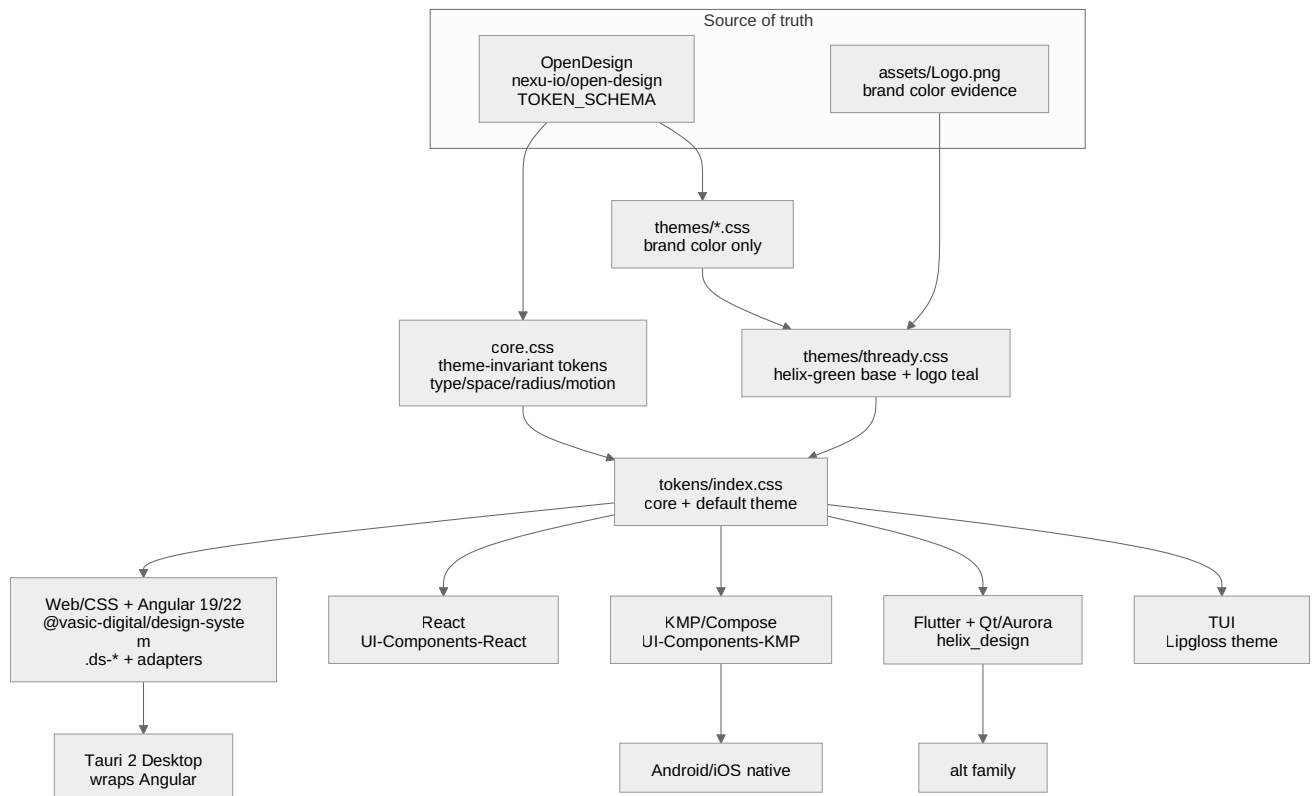
`[VERIFIED]` from `apps/daemon/src/brands/engine/export.ts` . **PenPot and Lottie are not native OpenDesign export targets** — see `prototypes.md` and `[OPEN: THREADY-DES-02]` .

The shared `vasic-digital/design_system` package is the normalized output of that pipeline for the Helix web surfaces. Its layers (`[VERIFIED]` from the repo `manifest.json`):

Layer	Path	Notes
Theme-invariant core tokens	<code>tokens/core.css</code>	type scale, spacing, radius, elevation, focus, motion, layout — no brand color
Brand themes (light + dark)	<code>tokens/themes/*.css</code>	<code>helix-green</code> (default), <code>vasic-red</code> , <code>helix-ota-blue</code>
Default entry	<code>tokens/index.css</code>	<code>core.css</code> + default green theme
Tailwind v4 layer	<code>tailwind/tailwind-v4.css</code>	token-bound utility layer
Fonts	<code>fonts/fonts.css</code>	Space Grotesk / Hanken Grotesk / JetBrains Mono (variable)
Universal components (CSS)	<code>components/css/components.css</code>	framework-agnostic <code>.ds-*</code>
Angular adapters	<code>components/angular/*</code>	<code>ThemeService</code> , <code>I18nService</code> , <code>ThemeToggle</code> , <code>LanguagePicker</code> , <code>DS_CONFIG</code>
i18n base	<code>i18n/en.json</code>	English shipped; +1 locale = 1 JSON + 1 <code>DS_LOCALES</code> row

Thready adds exactly one theme file (`tokens/themes/thready.css`), a Thready `DS_CONFIG` at app bootstrap, and the Thready-specific components that the shared `.ds-*` base does not yet cover (see `component-library.md`).

3. Token architecture



Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `diagrams/token-fan-out.mmd`.

Explanation (for readers/models that cannot see the diagram). Two inputs sit at the top: the OpenDesign `TOKEN_SCHEMA` (which fixes the *structural* tokens — type scale, spacing, radius, elevation, focus, motion, layout) and `assets/Logo.png` (which supplies the *brand color* evidence). OpenDesign produces two independent things: `core.css` (theme-invariant, carries no brand color) and the per-brand `themes/*.css` (brand color only). The Thready theme (`themes/thready.css`) is derived from the existing `helix-green` theme plus a teal secondary captured from the logo. `core.css` and the Thready theme combine in `tokens/index.css`, the single entry a consumer imports. From that entry the tokens fan out to every platform variant: the Web/CSS + Angular package (which the Tauri 2 desktop client wraps directly), a React variant, a KMP/Compose variant (the source for the Android/iOS native clients), a Flutter+Qt/Aurora variant (`helix_design`, the alternative family), and a Lipgloss theme for the TUI. Because brand color is isolated in one file, a white-label swap (see `theming.md`) re-tints every platform without touching structure.

3.1 Theme/Invariant core tokens (verbatim)

Reproduced from `tokens/core.css` [VERIFIED — IN-HOUSE: `design_system`]. These are declared once and **never re-bound per theme**:

```

:root {
  /* Fonts */
  --font-display: "Space Grotesk Variable", ui-sans-serif, system-ui, sans-serif;
  --font-body: "Hanken Grotesk Variable", ui-sans-serif, system-ui, sans-serif;
  --font-mono: "JetBrains Mono", ui-monospace, "SF Mono", Menlo, monospace;

  /* Type scale (OpenDesign `default` AI-structure, Tailwind-aligned) */
  --text-xs: 12px; --text-sm: 14px; --text-base: 16px; --text-lg: 20px;
  --text-xl: 24px; --text-2xl: 32px; --text-3xl: 48px; --text-4xl: 64px;
  --leading-body: 1.5; --leading-tight: 1.2; --tracking-display: -0.01em;

  /* Spacing (4px base) */
  --space-1: 4px; --space-2: 8px; --space-3: 12px; --space-4: 16px;
  --space-5: 20px; --space-6: 24px; --space-8: 32px; --space-12: 48px;
  --section-y-desktop: 80px; --section-y-tablet: 48px; --section-y-phone: 32px;

  /* Radius */
  --radius-sm: 8px; --radius-md: 12px; --radius-lg: 16px; --radius-pill: 9999px;

  /* Elevation & accent formula tokens (reference per-theme --fg/--border/--accent) */
  --elev-flat: none;
  --elev-ring: 0 0 0 1px var(--border);
  --elev-raised: 0 2px 8px color-mix(in oklab, var(--fg), transparent 92%);
  --accent-hover: color-mix(in oklab, var(--accent), black 8%);
  --accent-active: color-mix(in oklab, var(--accent), black 14%);
  --focus-ring: 0 0 0 3px color-mix(in oklab, var(--accent), transparent 70%);

  /* Motion */
  --motion-fast: 150ms; --motion-base: 200ms; --ease-standard: cubic-bezier(0.2, 0, 0, 1);

  /* Layout */
  --container-max: 1200px; --container-gutter-desktop: 24px;
  --container-gutter-tablet: 16px; --container-gutter-phone: 12px;
}

```

3.2 The Thready brand theme

The Thready theme extends the **verified** `helix-green` theme. `helix-green`'s brand color `#B6E376` was captured as the eyedrop **mean of 1,625,855 green-dominant saturated pixels** of the Helix Development logo, and its accessible accent is pinned to WCAG-AA (`#446E12` = 6.03:1 on white for light; `#B6E376` = 13.56:1 on `#020817` for dark) [\[VERIFIED — design_system/docs/THEMES.md\]](#).

Thready's own `Logo.png` is a **spiral/thread mark in chartreuse-green flowing to a mint/teal** (see [brand-assets.md](#)). The Thready theme therefore keeps the AA-verified helix-green accents and adds a **secondary teal** for the second brand color the spiral introduces:

```

/* tokens/themes/thready.css — import AFTER tokens/core.css */
/* [DEFAULT — adjustable] — extends the VERIFIED helix-green theme */
:root {
  --theme-id: "thready";

  /* Brand identity (decorative — logo marks / one focal element, NOT body text) */
  --brand: #b6e376; /* helix-green base [VERIFIED] — 1.47:1 on white, decorative only */
  --brand-2: #abddc9; /* Thready teal/mint secondary [VERIFIED — eyedrop mean of assets/
    Logo.png, see §3.2] */
  --brand-ink:#0a0f04; /* readable ink on a --brand fill (13.15:1) [VERIFIED] */

  /* Surface / foreground / border — brand-neutral slate, carried from helix-green [VERIFIED] */
  --bg: #ffffff; --surface: #ffffff; --surface-warm: #f1f5f9;
  --fg: #020817; --muted: #475569; --border: #e2e8f0; --border-strong: #64748b;

  /* Accent — LIGHT (helix-green, AA text on white) [VERIFIED] */
  --accent: #446e12; --accent-on: #ffffff;

  /* Semantic — LIGHT (AA) [VERIFIED] */
  --success: #166534; --warn: #854d0e; --danger: #dc2626;

  /* Footer heart (see brand-assets.md) — brand accent by in-house precedent */
  --ds-heart: var(--accent); /* [OPEN: THREADY-DES-03] confirm vs. classic love-red */
}

@media (prefers-color-scheme: dark) {
  :root:not([data-theme="light"]) {
    --brand: #b6e376; --brand-2: #b7ebd6; --brand-ink: #0a0f04; /* brighter teal (Logo.png
      median) for dark surfaces */
    --bg: #020817; --surface: #020817; --surface-warm: #1e293b;
    --fg: #f8fafc; --muted: #94a3b8; --border: #1e293b; --border-strong: #64748b;
    --accent: #b6e376; --accent-on: #0a0f04; /* logo lime IS the dark accent — 13.56:1 */
    --success: #16a34a; --warn: #eab308; --danger: #ef4444;
    --ds-heart: var(--accent);
  }
}

:root[data-theme="dark"], .dark { /* same dark block, for explicit-choice + .dark class */
  --brand: #b6e376; --brand-2: #b7ebd6; --brand-ink: #0a0f04;
  --bg: #020817; --surface: #020817; --surface-warm: #1e293b;
  --fg: #f8fafc; --muted: #94a3b8; --border: #1e293b; --border-strong: #64748b;
  --accent: #b6e376; --accent-on: #0a0f04;
  --success: #16a34a; --warn: #eab308; --danger: #ef4444;
  --ds-heart: var(--accent);
}

```

Provenance honesty. `--accent`, `--accent-on`, surfaces, foreground, border and semantic tokens are the **verified** helix-green/slate values (measured, AA). The Thready `--brand-2` teal is now **captured**, not provisional `[VERIFIED — eyedrop of assets/Logo.png, closes THREADY-DES-01]`: applying the design-system provenance rule (§11.4.6 — the mean of a color-dominant pixel region, the same method that produced helix-green `#B6E376`), the **mint/teal region** of `Logo.png` (1916×1522, pixels where

`blue > red` and `green ≥ blue`, saturated, opaque — **n = 618,886**) has mean `#ABDDC9` (median `#B7EBD6`, brightest sample `#B8ECD7`). The light theme uses the mean `#ABDDC9`; dark uses the brighter median `#B7EBD6` so the mark holds on the dark surface. The same capture over the **green-dominant** region (n = 1,057,661) returns `#BAE448`, which corroborates the documented helix-green base (`#B6E376`, median `#BEE747`) and validates the method. The exact heuristic is reproducible and should be re-confirmed by the design-system's own eyedrop tool at integration, but the value is captured evidence, not a guess. The teal is **decorative** (spiral marks, gradients, illustration) and **MUST NOT** be used for body text on light surfaces without an AA-verified darkening — matching the `--brand` rule.

3.3 Import order and entry

```
/* Thready app entry (product web + desktop) */
@import "@vasic-digital/design-system/tokens/core.css";
@import "@vasic-digital/design-system/tokens/themes/thready.css"; /* Thready default */
@import "@vasic-digital/design-system/fonts/fonts.css";
@import "@vasic-digital/design-system/components/css/components.css";
```

At Angular bootstrap the Thready app provides the `DS_CONFIG` so the shared `ThemeService` / `I18nService` key their storage and defaults correctly `[VERIFIED — ds.config.ts interface]`:

```
import { DS_CONFIG, DS_LOCALES, DS_DICTIONARY } from '@vasic-digital/design-system';

export const appConfig = {
  providers: [
    { provide: DS_CONFIG, useValue: {
      storagePrefix: 'thready', // -> localStorage 'thready-theme' / 'thready-lang'
      defaultTheme: 'system', // light | dark | system [OPERATOR: light+dark, system default]
      defaultLocale: 'en',
    } },
    { provide: DS_LOCALES, useValue: [
      { code: 'en', label: 'English' },
      { code: 'ru', label: 'Русский' },
      { code: 'sr-Cyrl', label: 'Српски' }, // [OPERATOR: en/ru/sr-Cyrl] ($12)
    ] },
    { provide: DS_DICTIONARY, useValue: THREADY_I18N },
  ],
};
```

3.4 Shipped brand themes & adding a theme

The package ships **three** brand themes today `[VERIFIED — tokens/themes/{helix-green,vasic-red,helix-ota-blue}.css + docs/THEMES.md]`. They matter to Thready twice: `helix-green` is the base the Thready theme extends, and all three are the **precedent** the per-Account white-label follows (a theme = one `--brand` + an AA-pinned `--accent`, neutrals shared). Measured accents:

Theme	--brand	--accent light (on white)	--accent dark (on #020817)	Provenance
helix-green (default, Thready base)	#B6E376	#446E12 = 6.03:1	#B6E376 = 13.56:1	eyedrop mean of 1,625,855 logo pixels [VERIFIED]
vasic-red	#E11D2A	#B91C1C = 6.47:1	#F87171 = 7.23:1	△ brand is a PLACEHOLDER pending the vasic logo asset [VERIFIED] — THEMES.md
helix-ota-blue	#2563EB	#2563EB	#3B82F6	HelixOTA canonical shadcn blue [VERIFIED]

The neutral/semantic slate is **shared across all three** (`--bg #ffffff / #020817`, `--muted #475569 / #94A3B8`, `--danger #DC2626 / #EF4444`, ...) and is AA-tuned — a theme changes only `--brand / --accent (+ -on)`. The theme file also declares convenience aliases the components read: `--fg-2 (= --fg)`, `--meta (= --muted)`, `--border-soft (= --border)` [VERIFIED — helix-green.css].

Adding a theme (the exact shipped 4-step, applied to **thready)** [VERIFIED — THEMES.md "Adding a theme"]:

1. Copy `tokens/themes/helix-green.css` → `tokens/themes/thready.css`.
2. Set `--theme-id`, `--brand`, `--accent (+ -on)` for light **and** dark; keep the neutral slate unless the brand truly needs a different surface family.
3. Record the brand-color provenance + a **measured** accent contrast ratio in `docs/THEMES.md` (no invented values [CONSTITUTION §11.4.6]).
4. Register it in `manifest.json > themes[]`.

This is precisely the workable item `THREADY-DES-DS-01` under `[GAP: 8.1]` — the Thready theme is a new theme file added by this exact process, plus a Thready `DS_CONFIG`, nothing forked.

4. Typography

Three shipped variable faces `[VERIFIED — fonts/fonts.css]`, all self-hostable via `@fontsource` (no external CDN — matches the offline/private posture and CSP hygiene):

Role	Token	Face	Use
Display	<code>--font-display</code>	Space Grotesk Variable	Headings <code>h1-h3</code> , hero, <code>.ds-display</code>
Body	<code>--font-body</code>	Hanken Grotesk Variable	Body, UI, controls
Mono	<code>--font-mono</code>	JetBrains Mono	Code, post payloads, CLI transcripts, hashtags, IDs

Type scale is the 8-step ramp in §3.1 (`--text-xs 12px ... --text-4xl 64px`). Body line-height is `1.5`; display is `1.2` with `-0.01em` tracking. All three faces are **variable** and self-hosted via `@fontsource` (no external CDN — matches the offline/private posture and CSP hygiene). Weight usage `[DEFAULT — adjustable]`: body/UI text 400–500 (`.ds-btn` is `font-weight:500` `[VERIFIED]`), headings 600–700 on `--font-display`, badges 600 `[VERIFIED — .ds-badge]`; the variable `wght` axis means these are subsets of one file, not extra downloads. Numeric/ID/hashtag content uses `--font-mono` with tabular figures so columns align in the Dashboard and Post-detail tables.

Localization is wired through the shipped `I18nService` (\$ component-library §4) — a `t(key)` read tracks the `lang` signal so switching re-renders instantly `[VERIFIED — i18n.service.ts]`. The footer slogan uses the exact keys `footer.made` / `footer.by` with `a11y.love` on the heart, and the language picker label uses `a11y.language` `[VERIFIED — reference.footer / language-picker component]`.

Cyrillic coverage (ru, `sr-Cyrl`) is required `[OPERATOR §12]`: verify each variable face’s Cyrillic subset at integration and fall back to the system stack per-glyph — tracked as `[OPEN: THREADY-DES-04]` (Cyrillic subset verification). Because logical properties and the token scale are direction-agnostic (§6.6), the type layer itself makes no LTR assumption.

5. Spacing, radius, elevation, motion

- **Spacing:** 4px base, ramp `--space-1...--space-12`; responsive section rhythm (`--section-y-{phone|tablet|desktop}` = 32/48/80px). Layout container `--container-max 1200px` with responsive gutters.
- **Radius:** `--radius-sm 8px` (controls), `--radius-md 12px` (cards), `--radius-lg 16px` (surfaces/sheets), `--radius-pill` (chips, toggles).
- **Elevation:** `--elev-ring` (1px token border), `--elev-raised` (soft shadow via `color-mix` over `--fg`). Prefer ring elevation in dark to avoid muddy shadows.

- **Motion:** two shipped durations — `--motion-fast 150ms` (hover/press, control state), `--motion-base 200ms` (enter/exit, disclosure) — on `--ease-standard cubic-bezier(0.2,0,0,1)`. Larger choreographed transitions (page/route, skeleton→content, processing animations) extend this in `prototypes.md` `[DEFAULT — adjustable]`. **All motion honors `prefers-reduced-motion: reduce`** — the shipped components already gate transitions on it `[VERIFIED — components.css]`.

6. Accessibility contract

Non-negotiable, aligned to the engineering-quality bar `[CONSTITUTION §11.4.190]` and the `design_system` ally evidence:

1. **WCAG 2.2 AA** minimum: text $\geq 4.5:1$ (normal) / $3:1$ (large + UI); non-text/UI boundaries $\geq 3:1$ (`--border-strong` exists precisely for load-bearing boundaries) `[VERIFIED]`.
2. **Brand vs. text separation:** `--brand` / `--brand-2` are decorative only (logo lime is 1.47:1 on white — never body text). Text uses `--accent` / `--fg`. **`--accent` (brand) and `--danger` (error) are separate tokens** — destructive UI is always `--danger`, so brand color never masks an error signal `[VERIFIED — THEMES.md]`.
3. **Focus visible everywhere:** `--focus-ring` (3px accent-tinted) on `:focus-visible`; the shipped `.ds-btn`, `.ds-input`, toggle and picker all implement it `[VERIFIED]`.
4. **Keyboard + SR:** every interactive component is reachable and operable by keyboard, with a correct accessible name (e.g. the theme toggle uses `aria-pressed`, the language picker is a native `<select>`) `[VERIFIED]`.
5. **Reduced motion** and **reduced transparency** respected.
6. **Localized, RTL-ready:** logical properties (`padding-inline`, `margin-inline`) are used in the shipped components; `en/ru/sr-Cyrl` are LTR, but the token layer does not assume direction.

Every one of these is a **testable gate** — see §8.

7. Per/platform fan/out

The design system fans to five platform variants (decision matrix §10.2). Honest status per `[GAP: 8.x]`:

Platform	Package / mechanism	Status	Thready plan
Web / CSS + Angular 19/22	@vasic-digital/design-system (.ds-*, adapters)	PRODUCTION -usable (web-only foundation) [GAP: 8.1]	Primary surface. Publish the package to npm; add Thready theme + Thready components. Web + CLI first [OPERATOR]
Desktop (Tauri 2)	Wraps the Angular web UI (Rust core)	Inherits web	No separate token work; OS-specific chrome overrides only
React	UI-Components-React	SCAFFOLD/ FLAGGED [GAP: 8.6]	Only if a React surface is needed (Catalogizer-style); re-audit before use
KMP / Compose (Android/iOS)	UI-Components-KMP + KMP fleet	SCAFFOLD, no CI/publish [GAP: 8.4]	Mobile shared-logic; needs CI + Maven publish + a token bridge (Compose Color / Dp from the CSS tokens)
Flutter + Qt/Aurora	helix_design	SCAFFOLD ("not yet implemented") [GAP: 8.2/8.3]	Alternative family; the only path to HarmonyOS/Aurora is native ArkTS/Qt via helix_shims [GAP: 8.5]
TUI	Lipgloss theme (Bubble Tea)	Pattern exists in-house [VERIFIED — helix_track/llms_verifier/.../tui]	Map the token palette to a Lipgloss Style set (see below)

Angular version split [Q17]. The Thready **product portal** — the primary management surface specified in `wireframes.md` — is **Angular 19** (matching the HelixTrack `web_client` + Tauri `desktop_client` family: Material 19, ngx-translate 17). **Angular 22** (standalone/signals, SSR + SSG/prerender, Tailwind v4 on OpenDesign tokens) is for **marketing / public sites**. Both consume the same `@vasic-digital/design-system` tokens + `.ds-*` set, which is why the Web platform row above is labelled “19/22”. The design system layer itself is version-agnostic.

Token bridging (the mechanism that keeps them one system). The CSS custom properties are the canonical source; each non-web platform consumes a **generated** binding, not a hand-kept copy, so a token change propagates. A token export step (OpenDesign `tokensToJson` → per-platform codegen) produces:

```
// TUI (Lipgloss) — generated from tokens/themes/thready.css [DEFAULT — adjustable]
package theme

import "github.com/charmbracelet/lipgloss"

// Thready dark palette (TUI defaults to the terminal's dark surface).
var (
    Accent    = lipgloss.Color("#B6E376") // --accent (dark)
    Fg        = lipgloss.Color("#F8FAFC") // --fg
    Muted     = lipgloss.Color("#94A3B8") // --muted
    Border    = lipgloss.Color("#1E293B") // --border
    Danger    = lipgloss.Color("#EF4444") // --danger
    Success   = lipgloss.Color("#16A34A") // --success
)

var Button = lipgloss.NewStyle().
    Foreground(lipgloss.Color("#0A0F04")). // --accent-on
    Background(Accent).
    Padding(0, 2).
    Bold(true)
```

```
// KMP/Compose — generated from the same tokens [DEFAULT — adjustable] [GAP: 8.4]
object ThreadyColors {
    val AccentLight = Color(0xFF446E12); val AccentDark = Color(0xFFB6E376)
    val Fg = Color(0xFF020817);          val FgDark = Color(0xFFF8FAFC)
    val Danger = Color(0xFFDC2626)
}

val ThreadySpacing = object { val s1 = 4.dp; val s4 = 16.dp; val s6 = 24.dp }
```

8. Visual/Regression & a11y testing

Mandated by [CONSTITUTION §11.4.162] (visual-regression required) and §10.2. The in-house family (Panoptic, VisualRegression (LLM-vision), ScreenDiff (pixel), ReplayBuffer) exists but is **library-grade with no CI** [GAP: 9.3] — the plan closes that:

Test type	Tool	Gate
Pixel visual-regression	<code>ScreenDiff</code>	Every component in every theme×state; fail on > threshold delta
Semantic visual-regression	<code>VisualRegression</code> (LLM-vision)	Catches layout/semantic breaks a pixel diff misses
Interaction replay	<code>ReplayBuffer</code>	Records + replays UX interaction sequences
a11y (web)	<code>cypress-axe</code> / Playwright + axe	Zero critical violations per screen <code>[VERIFIED — §9.4 test frameworks]</code>
Contrast tokens	scripted ratio check	Re-assert the THEMES.md ratios on every theme change
Challenges (scenario banks)	<code>vasic-digital/challenges</code>	Adversarial UI/UX decks (edge cases, overflow, RTL/Cyrillic, boundary contrasts) — mandated test type <code>[CONSTITUTION §11.4.27]</code> ; second half of <code>[GAP: 9.3]</code>

```
# thready visual-regression bank (HelixQA-style, anti-bluff runtime evidence)
suite: design-system.components
matrix:
  theme: [thready-light, thready-dark]
  state: [default, hover, focus-visible, active, disabled, error]
component: [ds-btn--primary, ds-btn--secondary, ds-input, ds-card, ds-nav, ds-footer, ds-badge]
evidence: [screenshot, dom-snapshot, axe-report] # runtime evidence mandatory (no bluff)
gate:
  pixel_delta_max: 0.1%
  axe_critical: 0
  contrast_min_text: 4.5
```

9. Gaps addressed & tracked workable items

- `[GAP: 8.1 design_system]` — **Plan:** add `tokens/themes/thready.css` (§3.2) upstream; publish `@vasic-digital/design-system` to npm; mature the standalone package. **Workable item** **THREADY-DES-DS-01.** Until published, consume via the git dependency. Do not claim non-web arms work.
- `[GAP: 8.2 helix_design]` / `[GAP: 8.3 helix_ui]` — **Plan:** implement the per-platform token packages (Flutter, Qt/Aurora, CSS) from the OpenDesign source, in lockstep with `design_system`,

before any Flutter/Qt client relies on them. Native ArkTS/Qt clients + `helix_shims` are the only HarmonyOS/Aurora path `[GAP: 8.5]`. **Workable items THREADY-DES-FLUT-01, -QT-01.**

- `[GAP: 8.4 UI-Components-KMP]` — **Plan:** add CI + Maven publish + a token-bridge codegen (§7) before the mobile clients depend on it. **Workable item THREADY-DES-KMP-01.**
- `[GAP: 9.3 VisualRegression family]` — **Plan (two halves):** (1) add CI to the visual-regression family and wire the bank in §8 — **workable item THREADY-DES-VR-01**; (2) author the Thready **Challenges** scenario banks (`vasic-digital/challenges`), a mandated test type) — **workable item THREADY-DES-CHAL-01** (see `component-library.md` §9).

10. Open items

- `[CLOSED: THREADY-DES-01]` Formal `Logo.png` eyedrop captured (§3.2): `--brand-2 = #ABDDC9` (light, mean of n=618,886 mint pixels) / `#B7EBD6` (dark, median). Re-confirm with the design-system eyedrop tool at integration.
 - `[OPEN: THREADY-DES-02]` PenPot + Lottie exports are not native to OpenDesign — bridge needed (see `prototypes.md`).
 - `[OPEN: THREADY-DES-03]` Heart-glyph color: brand accent (current default) vs. classic love-red.
 - `[OPEN: THREADY-DES-04]` Verify Cyrillic subsets of the three variable faces for ru / sr-Cyrl.
-

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